

Yuhan Liu

yuhanyh@bu.edu | [Google Scholar](#)

EDUCATION

Boston University , Boston, MA	Sep 2024-Present
Ph.D. in Mechanical Engineering (Advisor: Xin Zhang)	
Tsinghua University , China	Sep 2020-June 2023
M.S. in Electrical Engineering	
Xi'an Technological University , China	Sep 2015-Jul 2019
B.S. in Mechanical Engineering	

RESEARCH EXPERIENCES

Laboratory for Microsystems Technology , Boston University, Boston, MA	Sep 2024-Present
<i>Doctoral Student Researcher</i>	

Wireless RF Metamaterials for Magnetic Resonance Imaging

- Designed and validated a wireless circularly polarized volumetric RF metamaterial resonator at 127.74 MHz that improves MRI receive sensitivity, achieving $2.24\times$ SNR enhancement and reducing signal non-uniformity from 54.8% to 12.1% compared with a commercial multi-channel receive coil in a Philips 3T MR scanner.
- Developed and validated a wireless reconfigurable coaxial loop metasurface achieving up to $14.8\times$ SNR over a birdcage coil and $2.9\times$ as a clinical-array add-on at 3.0 T MR scanner, with demonstrated parallel-imaging compatibility.
- Built RF hardware prototypes through CST full-wave EM simulation, structural design, in-house PCB prototyping, VNA-based characterization, and 3T MRI testing.
- Implemented PIN-diode active detuning to decouple the wireless resonators during transmit, and verified the on/detuned states on a VNA; developed MATLAB workflows to process SNR maps, homogeneity metrics, VNA S-parameter data, and simulated field distributions.

Resonant Piezoelectric Energy Harvesting and Near-Field RF for Downhole Sensing (Advanced Energy Consortium)

- Built 3D multiphysics finite-element models (COMSOL) of resonant PZT bimorph cantilever energy harvesters, coupling structural, acoustic, and electrostatic domains to predict resonant modes and electromechanical response under acoustic-pressure excitation.
- Scaled a single cantilever into crossed dual-cantilever and 6-arm array configurations, raising the resonant frequency into 1-2 kHz and peak output voltage $3.6\times$ over the single-cantilever baseline.
- Showed that a cantilever length offset (ΔL) shifts each arm's resonant frequency and changes the combined output voltage, producing two independently tunable resonant peaks.
- Designed metamaterial-based near-field antennas in CST for wireless power transfer in lossy media, quantifying resonance shifts across materials of differing permittivity and conductivity.

Microelectromechanical Systems Lab , Tsinghua University, China	Sep 2020-May 2024
<i>Graduate Researcher</i>	

Wearable Electret Haptic Actuators and Emotional Interfaces

- Designed and fabricated multilayer PDMS electret actuators, reducing driving voltage from 500 V to 50 V while maintaining mechanical output; fabrication included multilayer PDMS casting with thickness control, cleanroom e-beam evaporation of Cr/Au electrodes (5/190 nm), and polyimide shadow-mask patterning to define an individually addressable 3×2 Braille-format array.
- Characterized actuators on a custom test bench (signal generator, high-voltage amplifier, oscilloscope, force sensor) with 50-450 Hz frequency sweeps and MATLAB signal processing (band-pass filtering, envelope extraction) to extract vibration output force.
- Developed a haptic emotion-encoding framework that separates the drive parameters into two perceptual axes—rhythm (frequency, vibration duration, pause time) governing arousal, and amplitude smoothness (amplitude level and its rate of change) governing pleasantness—and recombines them to place four fundamental emotions on a pleasantness-arousal map.
- Validated the framework with 20+ participants across user studies: 64.6% emotion-recognition accuracy without training, rising to 91.7-100% after a short learning phase, and 95.8% average accuracy in a navigation application.
- Provided the electret haptic hardware and device support for a digital cultural-heritage accessibility study, and contributed experimental work to collaborative studies on flexible electret actuators and piezoelectret sensors.

PUBLICATIONS

FIRST-AUTHOR PUBLICATIONS

Journals

- [1] **Y. Liu***, X. Zhu, K. Wu, S. W. Anderson, and X. Zhang. "Circularly Polarized Metamaterial Cage for Homogeneous Signal-to-Noise Ratio Enhancement in Magnetic Resonance Imaging." *Advanced Materials*, e16569, 2026.
- [2] **Y. Liu***, L. Han, S. Lv, T. Jiang, M. Duan, H. Guo, Y. Li, Q. Xie, Y. Chen, D. Wang, Z. Liu, W. Zhang, Y. Gong, J. Zhong, and X. Qian. "Emotional and Directional Enabled Programmable Flexible Haptic Interface for Enhanced Cognition in Disabled Community." *Research*, 8, 0714, 2025.

Conferences

- [3] **Y. Liu***. "Navigational and Emotional 4D Haptic Feedback Interface Based on Ultra-low Voltage Driven Electret Actuators." MRS Fall Meeting, Poster Presentation, Boston, USA, 2024.
- [4] **Y. Liu***. "Advanced Mental Health Monitoring with Modular Pixel Hydrogel Skin - Real-time Emotional and Stress State Management." MRS Fall Meeting, Poster Presentation, Boston, USA, 2024.

CO-AUTHOR PUBLICATIONS

Journals

- [5] T. Jiang, W. Qiu, Z. Li, X. Ye, **Y. Liu***, Y. Li, X. Wang, J. Zhong, X. Qian, and L. Lin. "Programmable Tactile Feedback Patterns for Cognitive Assistance by Flexible Electret Actuators." *Advanced Functional Materials*, 32(4), 2107985, 2022.
- [6] Y. Chen, Y. Liu, J. Liu, Y. Li, **Y. Liu***, W. Zhang, L. Han, D. Wang, S. Cao, H. Liu, H. Liu, Q. Xie, X. Wang, and M. Zhang. "Porous PDMS-ZnO Wearable Gas Sensor for Acetone Biomarker Detection and Breath Analysis." *ACS Applied Materials & Interfaces*, 16(45), 62436-62445, 2024.
- [7] Q. Xie, L. Han, J. Liu, W. Zhang, L. Zhao, **Y. Liu***, Y. Chen, Y. Li, Q. Zhou, Y. Li, Q. Zhou, Y. Dong, and X. Wang. "Kirigami-Inspired Stretchable Piezoelectret Sensor for Analysis and Assessment of Parkinson's Tremor." *Advanced Healthcare Materials*, 14(1), 2402010, 2025.
- [8] L. Han, **Y. Liu***, Z. Qi, Y. Zhao, N. Zhang, Y. Dong, and X. Wang. "Highly Polarized Piezoelectret Embedded Versatile Haptic Devices for Immersive Human-Robot Interactions." *Nano Energy*, 136, 110775, 2025.
- [9] L. Han, W. Liang, **Y. Liu***, W. Zeng, J. Wang, Z. Yang, Q. Zhou, Y. Dong, and X. Wang. "Stretchable Piezoelectret Electronic Stethoscope for Phonocardiography and Lung Sound Detection in Motion and Noise Conditions." *Applied Materials Today*, 36, 102077, 2024.
- [10] Y. Li, Y. Wang, Y. Liu, **Y. Liu***, L. Han, Y. Chen, J. Qu, P. Lai, and X. Qian. "Simultaneously Encapsulation and Formation of PDMS-MWCNTs Composites for Multidirectional Microchannel Force Sensors." *IEEE Sensors Journal*, 24(21), 34107-34116, 2024.

Conferences

- [11] K. Z. Zhou, W. Peng, **Y. Liu***, and R. F. Adler. "Exploring the Diversity of Music Experiences for Deaf and Hard of Hearing People." 28th ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW), 2024.
- [12] Y. Luo, W. Lin, **Y. Liu***, X. Nie, X. Qian, and H. Guo. "WeSee: Digital Cultural Heritage Interpretation for Blind and Low Vision People." IFIP Conference on Human-Computer Interaction, 123-131, 2023.

Manuscripts Under Review/In progress

- [13] **Y. Liu***, X. Zhu, K. Wu, K. Kaliaev, C. A. LeBedis, S. W. Anderson, and X. Zhang. A Wireless Reconfigurable Metasurface for Enhanced Parallel Magnetic Resonance Imaging. *Advanced Healthcare Materials*. *Under review. arXiv:2605.24791*.
- [14] **Y. Liu***, X. Zhu, C. A. LeBedis, S. W. Anderson, and X. Zhang. A Wireless Volumetric Metamaterial Resonator for Breast MRI. *arXiv preprint arXiv:2607.26360*, 2026.

AWARDS & HONORS

- Outstanding Student Leaders, Tsinghua University Oct 2022
- First Prize Academic Excellence, Tsinghua SIGS (top 10%) Oct 2021
- National Gold, 4th Youth Innovation and Entrepreneurship Competition, Youth Innovation and Entrepreneurship Competition Organizing Committee Dec 2020
- First Prize Academic Excellence, Xi'an Technological University (top 0.5%) Sep 2018

- Provincial First Prize, Shaanxi University Students Optoelectronic Design Competition, Shaanxi Province Optical Society Dec 2017

TEACHING & SERVICE

Teaching Assistant, Boston University

Fall 2025

ME310, Instrumentation

Student Union of Tsinghua Shenzhen International Graduate School (SIGS)

Sep 2021-Sep 2022

Vice President

TECHNICAL SKILLS

Computational Electromagnetics and Multiphysics Simulation: CST (full-wave EM), COMSOL Multiphysics (piezoelectric FEA: structural/acoustic/electrostatic coupling), RF resonator and antenna design, MRI coil modeling.

RF Testing and Characterization: VNA S-parameter measurement, resonant-frequency tuning, Q extraction with an inductively coupled loop probe, PIN-diode switching / active detuning, frequency-response and sweep testing, mechanical and electrical characterization.

Device Fabrication and Prototyping: Cleanroom e-beam evaporation (Cr/Au thin-film metallization), polyimide shadow-mask patterning, multilayer PDMS casting, plasma treatment, corona charging and high-voltage polarization, in-house PCB prototyping (LPKF), laser cutting, 3D printing, SolidWorks.

Instrumentation, Programming, and Data Analysis: Signal generator, high-voltage amplifier, oscilloscope, force sensor; MATLAB signal processing and data analysis, Python, Arduino, LabVIEW.